

Class 10 Science - Chapter 4

Carbon and its Compounds (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Class 8-9 ki yeh cheezein pehle pakki karo, warna Ch4 samajh nahi aayega.

1. Atom, Electron, Shell, Valence electron (Class 9 base)

Atom me electron alag-alag shell (K,L,M..) me ghoomte hain.

Valence electron = sabse bahar wali shell ke electron (bonding inhi se).

Carbon (C) ka atomic number = 6. Electron config = 2,4.

Yaani C ke paas bahar 4 valence electron hain.

Octet rule: har atom apni bahari shell me 8 electron (ya H ke liye 2) chahta.

2. Covalent bond - sharing wala bond

Covalent bond = do atoms ke beech electron ka SHARING.

Single bond = 1 pair share (jaise H-H, C-H).

Double bond = 2 pair share (jaise O=O).

Triple bond = 3 pair share (jaise N#N).

Carbon 4 electron share karta -> 4 covalent bond banata (tetravalent).

3. Valency aur Carbon ki khaasiyat

Valency = combine karne ki capacity. Carbon ki valency = 4.

Carbon na 4 electron de sakta (bahut energy) na le sakta -> isliye SHARE karta.

Do special powers:

Catenation = carbon ka carbon ke saath lambi chain/ring banana.

Tetravalency = 4 bond banane ki capacity (4 alag atoms se jud sakta).

4. Element, Compound, Molecule + formula bhasha

Molecular formula = molecule me atoms ki ginti (CH₄, C₂H₆, CO₂).

Hydrocarbon = sirf Hydrogen + Carbon se bana compound.

Reactant ----> Product, states (s)(l)(g)(aq). Equation balance.

5. Acid/Base/Salt (Ch2 link - jaroori for reactions)

Acid: khatta, blue litmus red. Base: kadwa, red litmus blue.

Acid + Base -> Salt + Water (neutralisation).

Combustion = oxygen me jalna (heat+light deta).

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

Class 10 Science | Ch 4: Carbon and its Compounds

[GREEN = board exam me baar-baar poocha jaata hai]

1. Bonding in Carbon - Covalent Bond

Carbon ke compounds me covalent bond hota (electron sharing).

Covalent compounds ke properties:

- Low melting/boiling point (kamzor inter-molecular force).
- Bijli ke kuchalak (poor conductor) - kyunki ions/free electron nahi.
- Aam taur par paani me kam ghulte (organic solvent me ghulte).

2. Carbon ki 2 super-powers [VERY IMPORTANT]

(1) CATENATION: carbon-carbon bond se lambi chains, branches aur rings.

Isi wajah se carbon ke laakhon (millions) compounds hain.

(2) TETRAVALENCY: carbon ke 4 valence electron $\rightarrow$ 4 bonds (C, H, O, N, S, Cl sab se jud sakta).

In dono ke kaaran carbon ke itne saare compounds bante hain.

3. Saturated vs Unsaturated [IMPORTANT]

Saturated hydrocarbon (ALKANES): sirf SINGLE bond. General C_nH_{2n+2} .

Eg: CH_4 (methane), C_2H_6 (ethane), C_3H_8 (propane).

Jalne par SAAF (blue) flame.

Unsaturated hydrocarbon: double/triple bond hota.

ALKENES (double bond) C_nH_{2n} : C_2H_4 (ethene).

ALKYNES (triple bond) C_nH_{2n-2} : C_2H_2 (ethyne/acetylene).

Jalne par DHUAAN-daar (yellow, sooty) flame (zyada carbon).

Unsaturated zyada reactive hote (addition reaction dete).

4. Chains, Branches, Rings + Homologous Series

Carbon compounds 3 shakal me: straight chain, branched chain, ring (cyclic).

HOMOLOGOUS SERIES = ek hi general formula wale compounds ka group, jisme har agla member $-CH_2-$ (14 mass unit) se zyada hota.

Eg alkane series: CH_4 , C_2H_6 , C_3H_8 , C_4H_{10} ...

Same chemical properties, dheere-dheere badalti physical properties.

5. Functional Groups [VERY IMPORTANT - RATTA]

Functional group = atom/group jo compound ko uska KHAAS property deta.

Group	Naam	Example
-OH	Alcohol	CH_3CH_2OH (ethanol)

-CHO	Aldehyde	CH ₃ CHO
>C=O	Ketone	CH ₃ COCH ₃
-COOH	Carboxylic acid	CH ₃ COOH (acetic acid)
-X (Cl, Br)	Halogen	CH ₃ Cl

Yaad: "Aldehyde -CHO, Ketone C=O beech me, Acid -COOH".

6. Nomenclature (Naam dena) - basics

Carbon ginti ke prefix: 1-Meth, 2-Eth, 3-Prop, 4-But, 5-Pent.

Saturated (single) -> "-ane" (methane). Functional group ka suffix lagta.

Eg: 2 carbon + -OH -> Ethanol. 2 carbon + -COOH -> Ethanoic acid.

7. Chemical Properties of Carbon Compounds [TOP EXAM AREA]

(A) COMBUSTION (jalna): Carbon/compounds O₂ me jalkar CO₂ + H₂O + heat+light.

CH₄ + 2O₂ -> CO₂ + 2H₂O + energy.

Saturated -> saaf blue flame; Unsaturated -> sooty yellow flame.

(B) OXIDATION: Alcohol ko oxidise karne par carboxylic acid banta.

CH₃CH₂OH --[O]/alkaline KMnO₄ ya K₂Cr₂O₇--> CH₃COOH.

Alkaline KMnO₄ / acidified K₂Cr₂O₇ = OXIDISING AGENTS.

(C) ADDITION REACTION (sirf unsaturated): H₂ add hota (Ni catalyst ke saath).

Vegetable oil + H₂ --Ni--> Vanaspati ghee (HYDROGENATION).

Oil (unsaturated) ko solid fat (saturated) banana.

(D) SUBSTITUTION REACTION (saturated/alkanes): H ki jagah doosra atom.

CH₄ + Cl₂ --sunlight--> CH₃Cl + HCl (ek-ek H, Cl se replace).

8. Ethanol (CH₃CH₂OH) [IMPORTANT]

Properties: liquid, paani me ghulta, alcoholic drinks me hota.

Reactions:

- Sodium ke saath: 2Na + 2CH₃CH₂OH -> 2CH₃CH₂ONa + H₂ (gas).

- Dehydration: conc. H₂SO₄ (170C) -> ethene (C₂H₄) + H₂O (paani hatta).

Misuse/effect: zyada alcohol nuksaandeh; methanol (CH₃OH) JAHREELA (andhapan/maut).

Denatured alcohol = ethanol + zaharila padarth (peene layak na rahe, sirf use).

9. Ethanoic Acid / Acetic Acid (CH₃COOH) [VERY IMPORTANT]

Acetic acid ka 5-8% solution = VINEGAR (sirka).

Glacial acetic acid = pure acetic acid (16-17C par ice jaisa jam jaata).

Reactions:

- Base ke saath: CH₃COOH + NaOH -> CH₃COONa + H₂O (neutralisation).

- Carbonate: 2CH₃COOH + Na₂CO₃ -> 2CH₃COONa + H₂O + CO₂ (effervescence).

- ESTERIFICATION: acid + alcohol --(conc.H₂SO₄)--> ESTER + water.

CH₃COOH + CH₃CH₂OH -> CH₃COOC₂H₅ (ester, meethi/fruity smell) + H₂O.

Ester se perfume + flavour bante. (NaOH se ulta -> saponification.)

10. Soaps and Detergents [IMPORTANT]

Soap = long-chain fatty acid ka sodium/potassium salt.

Saponification = fat/oil + NaOH $\rightarrow$ soap + glycerol.

Soap ka molecule: ek sira HYDROPHILIC (paani-pasand, ionic head),
doosra sira HYDROPHOBIC (paani-virodhi, carbon tail - mail/oil me ghusta).

MICELLE = mail ke around soap ke molecule ka gola (tail andar, head bahar) -
isi se mail paani me ghul kar dhul jaata.

Soap HARD water (Ca/Mg waala) me jhaag nahi banata $\rightarrow$ "SCUM" (chip-chipa)
banta. DETERGENT hard water me bhi kaam karta (isiliye behtar for washing).

11. Quick Revision - One Liners

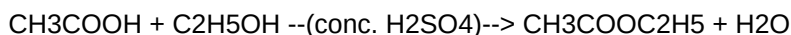
- Carbon: covalent, tetravalent, catenation $\rightarrow$ millions of compounds.
- Saturated=single (alkane, blue flame); Unsaturated=double/triple (sooty).
- Functional group property deta: -OH, -CHO, $>\text{C}=\text{O}$, -COOH, -X.
- Hydrogenation: oil + $\text{H}_2 \rightarrow$ ghee (Ni). Esterification: acid+alcohol $\rightarrow$ ester.
- Soap micelle se mail saaf; detergent hard water me bhi chalta.

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Esterification full + reverse

Q: Ethanol aur ethanoic acid ke beech reaction likho. Product ka naam, condition, smell? Reverse reaction kya kehlata hai?

Solution:



- Product: ester (ethyl ethanoate/ethyl acetate), meethi-fruity smell.
- Condition: conc. H_2SO_4 (catalyst + dehydrating agent), garam.
- Reverse (ester + NaOH -> alcohol + sodium salt) = SAPONIFICATION (soap banta).

Example 2 - Saturated vs unsaturated + flame + test

Q: Ethane aur ethene me antar? Flame se kaise pehchaan? Konsa addition deta?

Solution:

- Ethane (C_2H_6) saturated, single bond, SAAF blue flame.
- Ethene (C_2H_4) unsaturated, double bond, SOOTY yellow flame.
- Test: ethene (unsaturated) bromine water (orange) ka rang udaa deta (addition se), ethane nahi.
- Addition reaction sirf unsaturated (ethene) deta.

Example 3 - Why so many carbon compounds?

Q: Carbon ke itne saare compounds kyun hain? 2 reasons.

Solution:

- (1) Catenation - C-C lambi chain/ring banata.
 - (2) Tetravalency - 4 bonds (alag-alag elements se jud sakta).
- Iske saath chhota size -> strong stable bonds.

Example 4 - Soap kaise mail saaf karta

Q: Soap ka molecule mail (oil/grease) kaise hatata hai? Micelle samjhao.

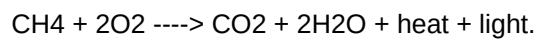
Solution:

- Soap ke do sire: hydrophilic head (paani me) + hydrophobic tail (oil me).
- Tail mail me ghuste, head paani ki taraf -> MICELLE ban jaata (gola).
- Mail ka gola paani me suspend hokar dhul jaata.

Example 5 (EASIEST) - Combustion equation

Q: Methane (CH_4) jalne ka balanced equation likho.

Solution:



PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Catenation aur tetravalency ka matlab.
- Q2. Saturated aur unsaturated hydrocarbon me 1 antar.
- Q3. Vinegar kya hai (kis acid ka kitna % solution)?
- Q4. Homologous series kya hai?

MEDIUM (2-3 marks)

- Q5. 5 functional groups uske naam ke saath likho.
- Q6. Hydrogenation kya hai? Equation/use (ghee).
- Q7. Saturated aur unsaturated ko flame/bromine water se kaise pehchaane?
- Q8. Soap aur detergent me antar (hard water ke context me).

HARD (3-5 marks, board favourite)

- Q9. Esterification reaction - equation, condition, product, reverse process.
- Q10. Ethanol ke 2 reactions (Na + dehydration) equation sahit.
- Q11. Substitution vs addition reaction - example sahit antar.
- Q12. Soap se safai - micelle ki madad se explain karo; hard water me kya hota?

ANSWER HINTS (PART D)

- A1. Catenation: C-C chain/ring; Tetravalency: 4 bond banana.
- A2. Saturated single bond (alkane); unsaturated double/triple (alkene/alkyne).
- A3. Vinegar = acetic acid ka 5-8% solution.
- A4. Same general formula wale compounds ka group, har agla -CH₂- jyada.
- A5. -OH alcohol, -CHO aldehyde, >C=O ketone, -COOH acid, -X halogen.
- A6. Oil + H₂ --Ni--> ghee; unsaturated ko saturated banana.
- A7. Saturated=blue flame; unsaturated=sooty + bromine water decolourise.
- A8. Detergent hard water me bhi jhaag deta; soap scum banata.
- A9. CH₃COOH + C₂H₅OH --H₂SO₄--> ester + H₂O; reverse = saponification.
- A10. 2Na+2C₂H₅OH->2C₂H₅ONa+H₂ ; conc.H₂SO₄ 170C -> ethene + H₂O.
- A11. Substitution (alkane, CH₄+Cl₂ sunlight); addition (alkene + H₂ / Br₂).
- A12. Head paani me, tail oil me -> micelle; hard water me soap scum.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

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